Traceable Predictive Maintenance for NASA C-MAPSS

An end-to-end PHM application with calibrated RUL prediction, deterministic risk logic, and auditable scenarios

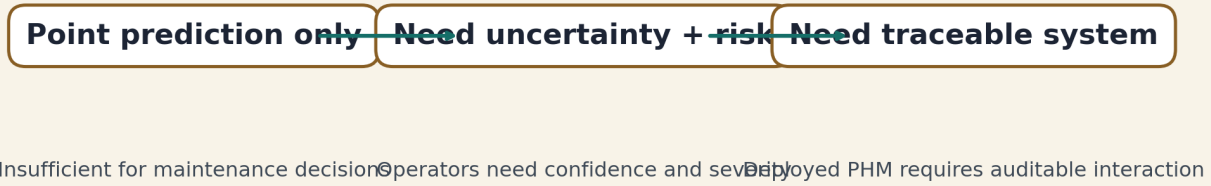
- Benchmark: NASA C-MAPSS
- System: Predictive Layer -> Agent Layer -> Dashboard Layer
- Focus: deployable PHM decision support rather than benchmark scores alone

Motivation and Problem

Why point RUL prediction alone is not enough

- Maintenance decisions require uncertainty, severity, and actionability.
- Benchmark optimization does not automatically produce a usable PHM application.
- This project closes that gap with calibrated prediction, deterministic decision logic, and traceable interaction.

Why benchmark RUL models are not enough



NASA C-MAPSS Context

Run-to-failure trajectories across four benchmark subsets

- FD001-FD004 vary in operating conditions and fault complexity.
- Each unit is a multivariate cycle trajectory with operational settings and sensor measurements.
- The task is to estimate Remaining Useful Life before failure.

Project Objectives and Contributions

What this system adds beyond benchmark prediction

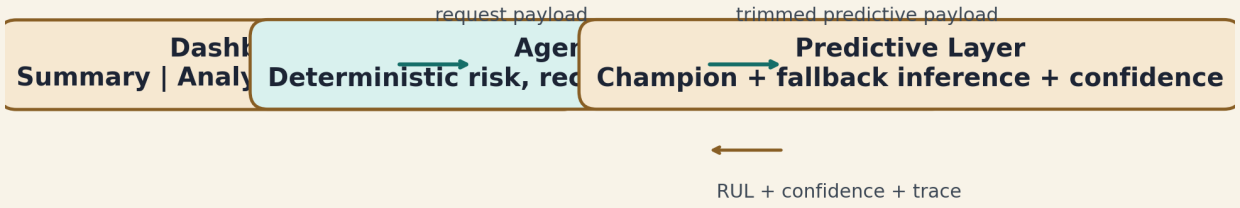
- A calibrated predictive layer with frozen preprocessing and champion selection.
- A deterministic agent layer for risk, recommendation, and scenario comparison.
- A dashboard with Summary, Analysis, Scenarios, and Technical Audit workflows.
- A reproducible artifact chain linking data, code, models, contracts, and audit traces.

System Architecture

Contract-driven Predictive -> Agent -> Dashboard stack

- The dashboard collects manual or CSV-driven requests.
- The agent layer trims payloads, computes deterministic risk and recommendations, and manages scenarios.
- The predictive layer serves the champion model with a calibrated confidence band and trace metadata.

Contract-based Predictive -> Agent -> Dashboard architecture



Data and Preprocessing

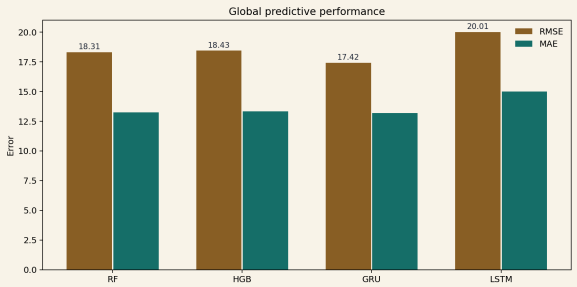
From raw C-MAPSS rows to model-ready features

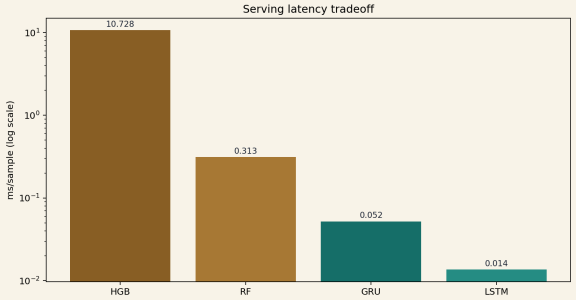
- The deployed predictive model uses 16 selected features after low-variance filtering.
- Training uses a training-only StandardScaler reused at serving time.
- The operational target is capped RUL with cap 125, while raw RUL remains available for audit and ablations.

Predictive Layer and Champion Selection

RF deployed despite GRU best raw RMSE (17.42 vs RF 18.31)

- Candidate models: RF, HGB, LSTM, and GRU.
- Champion: rf | Fallback: hgb.
- Selection reason: Best stable baseline selected for integration readiness.
- The deployed choice balances quality, stability spread, latency, and integration readiness.





Agent Layer and Deterministic Decision Logic

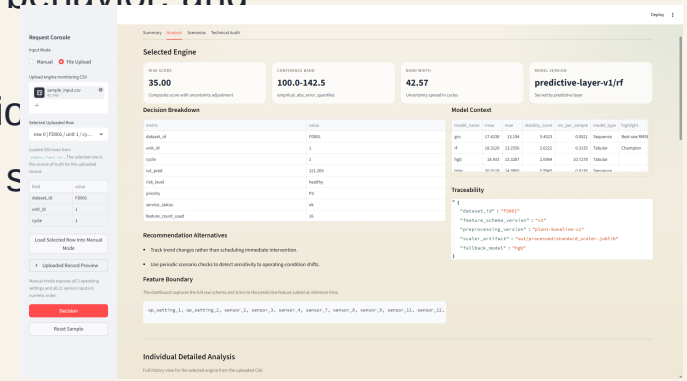
How RUL becomes risk, recommendations, and auditable scenario behavior

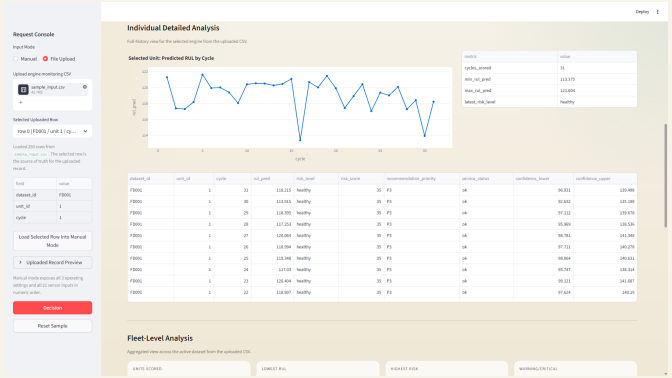
- Risk logic uses predicted RUL, confidence-band width, and service status.
- Recommendations are deterministic and aligned with the risk state.
- The LLM is optional and non-central: it may assist interpretation, but never controls execution or comparison values.

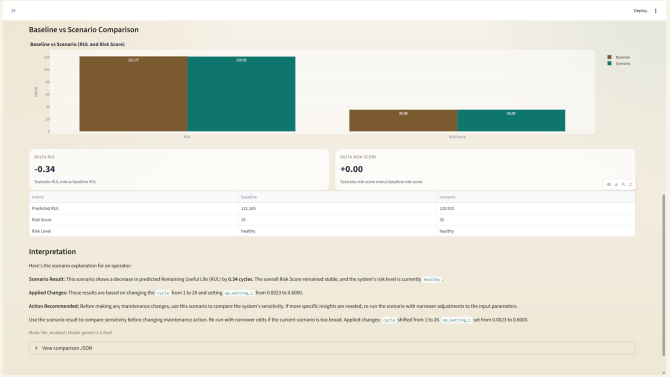
Dashboard Layer and User Workflow

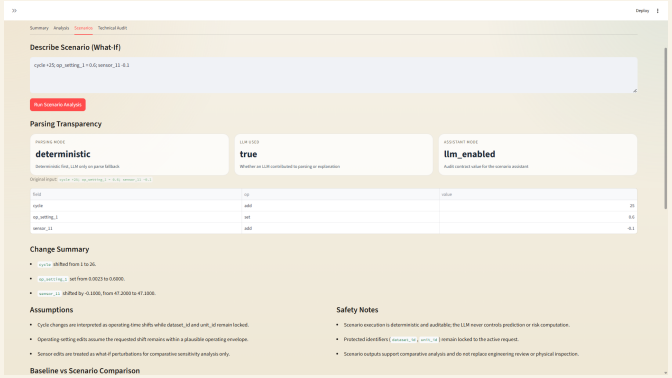
The UI operationalizes prediction, explanation, scenarios, and auditability

- Summary: decision-level RUL, risk, urgency, and action.
- Analysis: explanation, uncertainty, temporal behavior, and fleet context.
- Scenarios: deterministic baseline-vs-scenario
- Technical Audit: trace fields, model version, s and contracts.





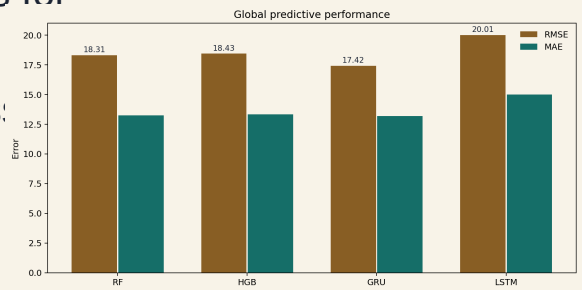


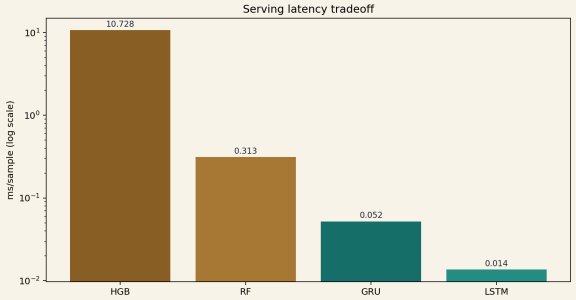


Experiments and Results

Evidence combines prediction quality, stability, and engineering tradeoffs

- Global metrics: GRU best raw RMSE; RF strongest stable tabular baseline.
- Per-dataset metrics show why stability spread matters for deployment.
- Latency and fallback behavior are treated as first-class engineering evidence.

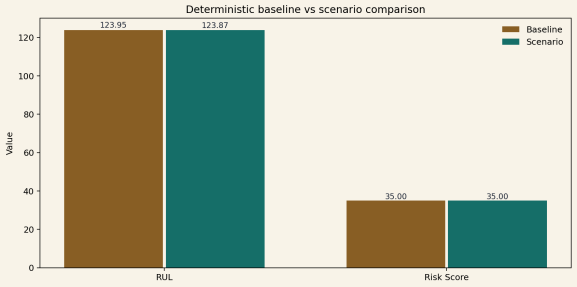


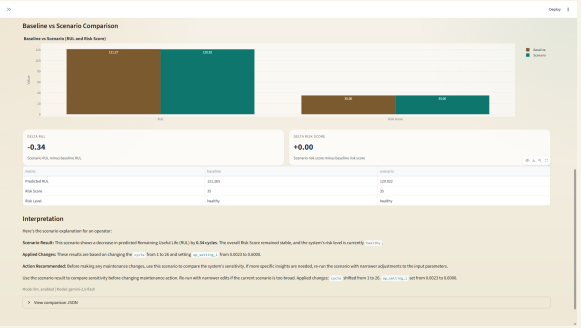


Scenarios and Interpretability

Deterministic what-if analysis with optional non-central LLM assistance

- Scenario parsing is deterministic-first and only uses the LLM as a fallback for intent interpretation.
- Example scenario: baseline vs scenario delta RUL = -0.081 cycles.
- Risk score remained 35 -> 35, showing how some edi local impact.
- This makes scenario outputs useful for comparative s analysis rather than speculative control.





Engineering Constraints

Deployment decisions were shaped by cost, latency, safety, reliability, and traceability

- Cost: the LLM remains optional and non-central.
- Latency: the deployed RF path is simple and fast to serve.
- Safety: validation rules and protected identifiers constrain scenario execution.
- Reliability: explicit fallback and service-state propagation are surfaced end-to-end.
- Traceability: audit records and contracts support debugging and review.

Limitations and Future Work

What the current system does well and where it still needs extension

- The deployed champion is a single-row tabular model rather than a temporal serving model.
- Some scenario edits have low local sensitivity because they target non-features or weak local directions.
- The confidence-band policy is useful but relatively simple.
- Future work: richer local explanation, broader scenario libraries, stronger fleet analytics, and live deployment experiments.

Conclusion

The project demonstrates a deployable and traceable PHM system, not only a benchmark model

- Scientifically: calibrated RUL modeling with documented champion selection.
- Engineering-wise: deterministic decision support, scenarios, and traceability integrated into one user-facing system.
- Main lesson: deployment quality depends on more than raw RMSE.

Backup / Appendix

Extra evidence for Q&A

- Full per-dataset metrics and latency table.
- Selected feature list and service-state taxonomy.
- Contract field mapping and reproducibility commands.
- Scenario parsing examples and failure-mode handling.